CASE STUDY 20 – UNIVERSITY ADMISSION DECISION ENGINE

1. Problem Statement

Design a Python program to evaluate university admission applicants based on subject-wise eligibility, aggregate marks, course-specific minimum thresholds, and predefined tie-breaking rules. The program identifies eligible applicants and ranks them according to their performance.

2. Objectives

1. Check minimum marks in all subjects.
2. Calculate aggregate percentage.
3. Apply course-specific minimum aggregate requirements.
4. Separate eligible and ineligible applicants.
5. Rank eligible applicants using predefined tie-breaking rules.

3. Assumptions

Three subjects are considered: Mathematics, Physics, and Computer Science. Minimum subject mark is 50. Course thresholds are CSE = 70%, ECE = 65%, EEE = 60%. Ties are resolved by higher Mathematics marks, then higher Computer Science marks, then alphabetical order.

4. Algorithm

1. Store applicant details in a list of tuples.
2. Store course thresholds.
3. Calculate total and aggregate for each applicant.
4. Check subject-wise minimum marks and course threshold.
5. Store applicants in eligible or ineligible lists.
6. Rank eligible applicants using nested loops and tie-breaking rules.
7. Display the final ranking and ineligible applicants.

5. Pseudocode

START
Set minimum subject mark = 50
Set course thresholds
Store applicant details
FOR each applicant: calculate total and aggregate; check subject eligibility and course threshold; store in eligible or ineligible list
FOR each pair of eligible applicants: compare aggregate, Mathematics, Computer Science and name; swap when required
Display ranking and ineligible applicants
STOP

6. Flowchart

START → Store applicant details → Set minimum subject mark and course thresholds → Calculate aggregate → Check subject eligibility → Check course threshold → Eligible? → YES: Eligible List / NO: Ineligible List → Rank eligible applicants → Apply tie-breaking rules → Display results → STOP

7. Python Source Code

```
# University Admission Decision Engine

cse = 70
ece = 65
eee = 60

minimum_subject_mark = 50

# (Name, Course, Mathematics, Physics, Computer Science)
```

```
applicants = [
    ("Anil", "CSE", 82, 76, 88),
    ("Bhavya", "ECE", 70, 68, 72),
    ("Charan", "CSE", 74, 70, 76),
    ("Divya", "EEE", 65, 60, 70),
    ("Esha", "CSE", 69, 72, 75),
    ("Farhan", "ECE", 70, 68, 72)
]

eligible = []
ineligible = []

for applicant in applicants:
    name, course, maths, physics, computer = applicant

    total = maths + physics + computer
    aggregate = total / 3

    if course == "CSE":
        course_threshold = cse
    elif course == "ECE":
        course_threshold = ece
    else:
        course_threshold = eee

    if (maths >= minimum_subject_mark and
        physics >= minimum_subject_mark and
        computer >= minimum_subject_mark and
        aggregate >= course_threshold):

        eligible.append(
            (name, course, maths, physics, computer, aggregate)
        )
    else:
        ineligible.append(
            (name, course, aggregate)
        )

# Ranking
for i in range(len(eligible)):
    for j in range(i + 1, len(eligible)):
        first = eligible[i]
        second = eligible[j]
        swap = False

        if first[5] < second[5]:
            swap = True
        elif first[5] == second[5] and first[2] < second[2]:
            swap = True
        elif (first[5] == second[5] and
              first[2] == second[2] and
              first[4] < second[4]):
            swap = True
        elif (first[5] == second[5] and
              first[2] == second[2] and
              first[4] == second[4] and
              first[0] > second[0]):
            swap = True

        if swap:
            eligible[i], eligible[j] = eligible[j], eligible[i]

print("UNIVERSITY ADMISSION RANKING")
print("----------------------------")

rank = 1

for applicant in eligible:
    name, course, maths, physics, computer, aggregate = applicant

    print("Rank:", rank)
    print("Name:", name)
    print("Course:", course)
    print("Mathematics:", maths)
    print("Physics:", physics)
    print("Computer Science:", computer)
    print("Aggregate:", round(aggregate, 2), "%")
    print("----------------------------")

    rank = rank + 1

print("INELIGIBLE APPLICANTS")
print("---------------------------")
```

```
for applicant in ineligible:
    name, course, aggregate = applicant

    print("Name:", name)
    print("Course:", course)
    print("Aggregate:", round(aggregate, 2), "%")
    print("----------------------------")
```

8. Test Cases

Test Case	Applicant	Course	Expected Result
1	Anil	CSE	Eligible
2	Bhavya	ECE	Eligible
3	Charan	CSE	Eligible
4	Divya	EEE	Eligible
5	Esha	CSE	Ineligible
6	Farhan	ECE	Eligible

9. Sample Output

```
UNIVERSITY ADMISSION RANKING
----------------------------
Rank: 1
Name: Anil
Course: CSE
Mathematics: 82
Physics: 76
Computer Science: 88
Aggregate: 82.0 %
----------------------------
Rank: 2
Name: Charan
Course: CSE
Mathematics: 74
Physics: 70
Computer Science: 76
Aggregate: 73.33 %
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Rank: 3
Name: Bhavya
Course: ECE
Mathematics: 70
Physics: 68
Computer Science: 72
Aggregate: 70.0 %
----------------------------
Rank: 4
Name: Farhan
Course: ECE
Mathematics: 70
Physics: 68
Computer Science: 72
Aggregate: 70.0 %
----------------------------
Rank: 5
Name: Divya
Course: EEE
Mathematics: 65
Physics: 60
Computer Science: 70
Aggregate: 65.0 %
----------------------------

INELIGIBLE APPLICANTS
----------------------------
Name: Esha
Course: CSE
Aggregate: 72.0 %
----------------------------
```

10. Conclusion

The University Admission Decision Engine evaluates applicants using subject-wise minimum marks and course-specific aggregate requirements. It separates eligible and ineligible applicants and ranks eligible

candidates using predefined tie-breaking rules. The program demonstrates variables, data types, operators, conditional statements, loops, lists, tuples, and tuple unpacking.